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ARTICLE

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Baited remote underwater video stations as a potential tool for assessing coastal rocky fishes in Chile

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Abstract

Coastal rocky fish species support important fishing activities, but basic information is lacking for many before catches decline. Monitoring and management of coastal rocky fish species is urgently needed in Chile, but knowledge of these species is fragmentary at best. Performance of Baited Remote Underwater Video Stations (BRUVS) for assessing rocky fish assemblages was evaluated in south-central Chile. Seven medium- to large-sized exploited species dominated fish assemblages in the survey area, including five small cryptic rocky reef fishes, one agnathan and a catshark species. These seven species represented 78% of the species targeted by commercial fishing in the survey area. Furthermore, this tool provided new information on behaviour of rocky fish species. The simple, inexpensive and light nature of single-camera BRUVS may be crucial for maintaining monitoring programs along remote areas of Chile.

KEYWORDS
BRUVS, fish monitoring, rocky reef fish assemblage, southeastern Pacific

1 | INTRODUCTION

Rocky reef fishes support important commercial and recreational fishing activities throughout the world. Given the small and local scales of these fisheries for local communities, basic information is lacking for many of them, such as species identification, catch and fishing effort. Worldwide, catches of coastal rocky fish species have declined over time (Johnson & Jackson, 2015). The cumulative effect of fishing effort and lack of basic fisheries and biological information contributed to the decline of coastal fish stocks (Gough et al., 2020). In Latin America, assessing and managing coastal fish species is challenging because information on catch and fishing effort is difficult for managers to monitor because effort is sparsely distributed and catches mainly supply local markets. Many coastal rocky fish species also exhibit particular biological attributes that make them highly vulnerable to fishing exploitation, such as high longevity, sequential hermaphroditism, late maturity and site fidelity or philanthropy (Godoy et al., 2010).

Chile lands approximately 2.1 million tons of fish every year, the tenth largest catch of any country (FAO, 2020). Despite the historical importance of industrial fishing in Chile, knowledge regarding subsistence fisheries based on coastal rocky fish species is fragmentary at best. These fisheries are not monitored in Chile, and catch and effort are not restricted or conserved. Some coastal rocky species in Chile are highly valuable in restaurants in major urban cities, and a decade ago they started to exhibit signs of overexploitation, such as decreased catches, reduced fish availability and truncation of size composition (Godoy et al., 2010).

Continental Chile includes more than 4000 km of coastline, and ~ 1050 coastal rocky teleost fishes have been identified (Pequeño & Sáez, 2010), displaying a clear biogeographic pattern (Ojeda et al., 2000). In Chile, diversity of coastal rocky fish

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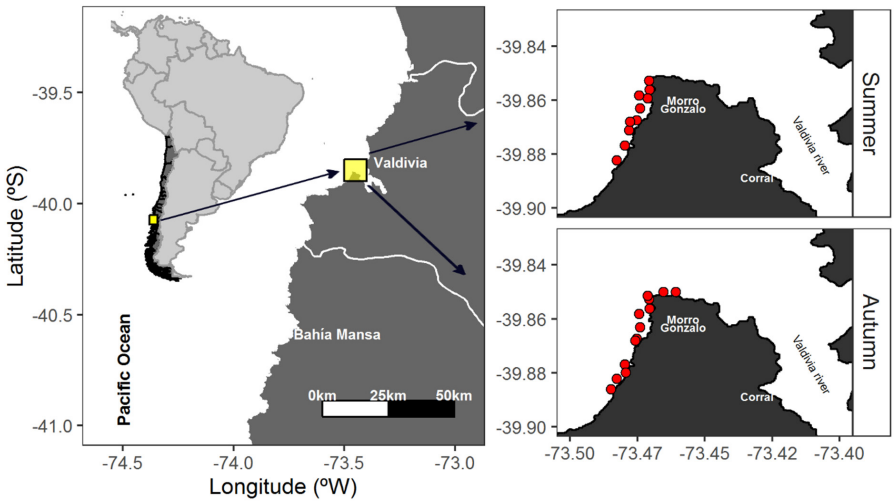

FIGURE 1 Study area showing sampling locations (Red solid dots) for coastal rocky fish species in Baited Remote Underwater Video Stations (BRUVS) at Morro Gonzalo in the Los Ríos region of south-central Chile on 16–17 January (summer) and 29–30 April (autumn) 2022.

species declines with latitude because of changes in the continental slope topography, substrate heterogeneity and oceanographic conditions (Angel & Ojeda, 2001; Ojeda et al., 2000). Currently, ~17 coastal rocky fish species are targeted by commercial and recreational fishing in Chile (The Nature Conservancy, 2020). The diversity of coastal fishes off Chile and the extension and shape of the coastline cause fishing communities to be isolated, so monitoring and managing these species is extremely challenging.

Fisheries management in Chile is based on landing records of the main species and fisheries–biological attributes recorded by the Instituto de Fomento Pesquero (IFOP). Well-established methods for assessing populations, such as acoustic techniques, are inapplicable in coastal environments, which adds complexity to assessing coastal marine fisheries. Longline surveys, an alternative for assessing fish populations in Chile (Wiff et al., 2020), are ineffective in coastal environments because of high current velocities and highly heterogeneous rocky substrate that produce drifts and entanglements of fishing gear. Traps are a common method of assessing shallow coastal fish species (e.g. Recksiek et al., 1991), but traps are almost exclusively used in Chile for demersal crustaceans, with only experimental use for fishes (Arancibia, 2001). Another common method to assess coastal fishes is underwater visual census techniques (Samoilys & Carlos, 2000). However, given the large expanse of coastal areas, remoteness and commonly rough weather conditions, this technique is not amenable for mass application and replication along the Chilean coastline. Therefore, an alternative method for assessing coastal fish species in Chile needs to be explored.

Baited remote underwater video stations (BRUVS) are based on a static video imagery collection of fishes that are attracted to bait or present within the field of view of the camera for a period of time (e.g. Merlo et al., 2023). This technique has grown exponentially in the last decade, mostly for the study of reef fish ecology (Mallet & Pelletier, 2014; Prat-Varela et al., 2023). However, use of BRUVS has diversified in recent years (e.g. Gladstone et al., 2012), and has now been adopted as a regular monitoring tool by fisheries agencies in developed countries (e.g. Thompson et al., 2022). BRUVS have been successfully used in Argentinian Patagonia to assess several fish species and benthic habitats, from small cryptic species to large shark species from coastal areas (Trobbiani et al., 2021), rocky reef fishes (Merlo et al., 2023), large shark aggregation areas (Irigoyen et al., 2018, 2019) and mesophotic and aphotic deep areas (Irigoyen, unpublished data). Together with the increase in the use of BRUVS, several research groups on fish ecology are comparing this technique with underwater visual census (UVC), a more classical methodology to assess reef fish assemblages. Although most studies agree on the complementarity of both techniques, BRUVS are more versatile for diver-shy species, use in areas beyond safe depths for divers (25 m), and monitoring fish assemblages over non-consolidated bottoms (with typically shy or low-density species) that are commonly not suitable for UVC (e.g. Gladstone et al., 2012; Trobbiani et al., 2021). Furthermore, trained or certified divers are not needed for setting BRUVS sampling stations, which makes this technique more amenable for substantial and standardised implementation. Despite many advances in BRUVS, drawbacks of the technique are related to the unknown area of inference due to variability of bait odour plumes and possible negative interactions among species attracted to bait (Taylor et al., 2013).

Our objective was to determine if BRUVS could be used to estimate relative abundance, species richness and diversity of coastal rocky fish species in Chile. In this region, experiments using BRUVS to assess coastal fish stocks were largely fragmented and highly experimental (Araya et al., 2015, 2018; Riquelme-Pérez et al., 2019). However, BRUVS could be a relatively inexpensive and easy technique to implement for assessing and monitoring coastal rocky fishes in Chile. Here, we evaluated performance of BRUVS for assessing, monitoring and identifying coastal rocky fish assemblages in Chile. We selected the coastal community off the Los Ríos region in south-central Chile (~40°S latitude, Figure 1) to assess coastal rocky fish assemblages and to provide a methodological base for how to assess and monitor temperate rocky fish species in Chile.

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2 | MATERIALS AND METHODS

2.1 | Study site

The study area, Morro Gonzalo, in the Los Ríos region in south-central Chile, is a well-known local fishing ground for coastal rocky fish species (Figure 1). Water surface temperature fluctuates seasonally between a mean value of 14.2°C in summer (January) and 9.5°C in winter (August) (https://es.seatemperature.net/current/chile/valdivia). However, water temperature is highly variable in this region at a short temporal scale. Field records of water temperature collected during this study with sensors included on Vaquita Paralenz® cameras varied from 9.1 to 9.4°C in January (summer) and 11.3 to 11.4°C in April (autumn).

2.2 | Sampling design and BRUVS

Sampling trips were on 16–17 January (summer) and 29–30 April (autumn) 2022. BRUVS consisted of iron poles and a trestle-shaped frame enclosing GoPro® (models Hero 3+ and Hero 4) Paralenz® (Vaquita) and SJCAM® (SJ8 Air HD) cameras. A perforated bait canister 110 mm diameter and 20 cm long, stuffed with 600 g of fresh chopped jack mackerel (*Trachurus murphyi*), was held with detachable 110 cm-long arms (25 mm Ø plastic conduit) (See Appendix S1). Of 29 BRUVS stations, 14 were randomly distributed in summer and 15 in autumn in the exposed area of Morro Gonzalo (39°52′S, 73°28′W) (Figure 1). BRUVS were deployed at depths of 5–25 m and times of 10:00 am–3:00 pm. Environmental conditions were similar with horizontal visibility of ~10 m (estimated by eye using as a reference the visible part of the BRUVS structure on video imagery) and slight current. BRUVS were set for 60–75 min of recording time and 400-m apart to ensure independence between samples. Positions of BRUVS were recorded using GPS. Depth and a priori bottom-type classification provided by local fishermen with an echosounder were cross-validated with depth and temperature recorded by Paralenz® cameras.

2.3 | Data analysis

For consistency in the identification of fish species, videos were analysed by the same researcher using Sony Vegas™ software. For each video record, 60 min of video imagery was analysed starting after stabilisation of the BRUVS on bottom. Each observed fish was identified to the lowest taxonomic level possible from the video image. For each taxon identified, relative abundance (MaxN) and time of first arrival (TFA) (Ellis & DeMartini, 1995) were calculated. MaxN estimated relative fish abundance as the highest number of individuals of each species within the field of view at the same time during the entire video footage (i.e. the best index of relative abundance from BRUVS; Cappo et al., 2004, 2006; Whitmarsh et al., 2017). TFA was the time in minutes from camera landing to the time at which the first individual of each taxon was detected, for habitat association interpretation of detected species and comparison of fish activity between sampling events (Irigoyen, personal observations). For each BRUVS deployed, species richness, the Shannon–Wiener diversity index (Begon et al., 2006), total abundance (the sum of the MaxN of each taxon recorded) and mean TFA (the average TFA including all taxa recorded in a deployment) were computed.

To ensure that changes in richness and abundance of fish species were not constrained by a small sampling size (Chao et al., 2014), rarefaction curves were constructed using first-order ($q = 0$, taxonomic richness) and second-order ($q = 1$, species diversity) Hill numbers (Chao et al., 2014; Colwell et al., 2012; Jost, 2007). A sampling coverage function curve for each season was also plotted using the iNEXT package of R software (Chao et al., 2014; Hsieh et al., 2020). Sampling coverage is a measure of sample completeness that reflects the relative proportion of taxa present in the habitat that were included in the sample (Chao & Jost, 2012). Each curve was extrapolated to the double sample size (number of reefs) and 95% confidence bands were estimated using a bootstrap method (100 replicates, Chao et al., 2014). Generalised linear models (GLMs) were used to evaluate differences in species richness (S), diversity (H′), mean time of first arrival (TFA), total abundance (MaxN) and mean abundance of the five most conspicuous species among the two sampling seasons. For GLMs, error distributions were Poisson for species, Gaussian for Shannon diversity and TFA, and negative binomial for the total number of observed fish (abundance). p values were corrected for multiple comparisons using the Bonferroni correction (Holm, 1979) with the "p.adjust" function of the "stats" package of R software (R Development Core Team, 2020).

3 | RESULTS

Of 23 BRUVS deployments, 10 were in summer and 13 were in autumn (Table 1). Deployments covered 5–25 m of depth over rocky bottoms at an average depth of 13.8 m in summer and 14.5 m in autumn (Table 1).

Sample coverage reached 90% for summer surveys and 94% for autumn surveys (Figure 2III). Of 14 fish taxa recorded, nine were in summer and 12 were in autumn, including 12 Osteichthyes, one Agnatha and one Chondrichthyes (Table 1). Five fish species accounted for 87.8% of total abundance (MaxN) (Figure 3), of which three were conspicuous (medium- to large-sized commercially relevant species) rocky reef fish species (*Prolatilus jugularis*, *Pinguipes chilensis* and *Sebastes oculatus*). One small cryptic rocky reef species (*Patagonotothen* sp.) and one scavenger (*Eptatretus* sp.) associated with mud and rocky habitats (Reyes & Hune, 2012) were also recorded. The other four taxa of medium to small rocky reef cryptic species accounted for 10.4% of total abundance (*Bovichtus chilensis*, *Calliclinus* sp., *Scartichthys viridis* and *Helcogrammoides cunninghami*). Five other taxa were recorded in only one BRUVS deployment,

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TABLE 1 Relative abundance (MaxN), species richness, mean time of first appearance (TFA), and depth of capture for 14 fish species recorded per Baited Remote Underwater Video Stations (BRUVS) deployment at Morro Gonzalo in the Los Ríos region of south-central Chile on 16–17 January (summer) and 29–30 April (autumn) 2022.

	N_{max}									
Sampling trip	**Summer**									
Taxa/BRUVS	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Pinguipedidae										
Blanquillo *Prolatilus jugularis*	3	3	6		1	2	1		1	
Rollizo *Pinguipes chilensis*	2	3	5	3	3					
Sebastidae		10		2	2	3	2	6	1	
Cabrilla *Sebastes oculatus*										
Nototheniidae	5		4	3	4					
Patagonotothen sp.										
Myxinidae			2	1	1					
Anguila babosa *Eptatretus* sp.										
Bovichtidae	1		1							
Torito *Bovichtus chilensis*										
Ophidiidae		1								
Congrio colorado *Genypterus chilensis*										
Labrisomidae				1						
Tromboyo *Calliclinus* sp.										
Aplodactylidae				1						
Jerguilla *Aplodactylus punctatus*										
Blenniidae										
Borrachilla *Scartichthys viridis*										
Scyliorhinidae										
Pintarroja *Schroederichthys* sp.										
Cheilodactylidae										
Bilagay *Cheilodactylus variegatus*										
Kyphosidae										
Mulata *Graus nigra*										
Tripterygiidae										
Trombolito *Helcogrammoides cunninghami*										
Total N_{max}	11	17	18	11	11	5	3	6	2	0
Species Richness	4	4	5	6	5	2	2	1	2	0
Time of first appearance	0:06:02	0:06:25	0:16:59	0:07:37	0:23:00	0:25:11	0:04:46	0:02:36	0:28:25	
Depth (m)	8	22	9.5	10	5	20	16	14	16	18

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Autumn													
11	12	13	14	15	16	17	18	19	20	21	22	23	
3	3	3	4	5	2		6	6	4		1	5	
4	1	1	3	1	4	2	5	4	2	3	2	4	
1		3	6	9	3					5		1	
5	2	4			8	2	3	3	1		3		
2	1	6	3	3	6	2	1	2	1	4	4		
								1		1	1	1	
1											2		
3		4			1		1	1	1		3		
					1								
							1						
										1			
										3	2		
19	7	21	16	18	25	6	17	17	9	17	18	11	
7	4	6	4	4	7	3	6	6	5	6	8	4	
0:07:29	0:06:07	0:16:50	0:03:17	0:04:11	0:15:53	0:17:38	0:19:20	0:12:01	0:13:51	0:09:14	0:17:37	0:07:52	0:03:03
12	21	13	20	16	25	17	15	12	11	11	10	7	

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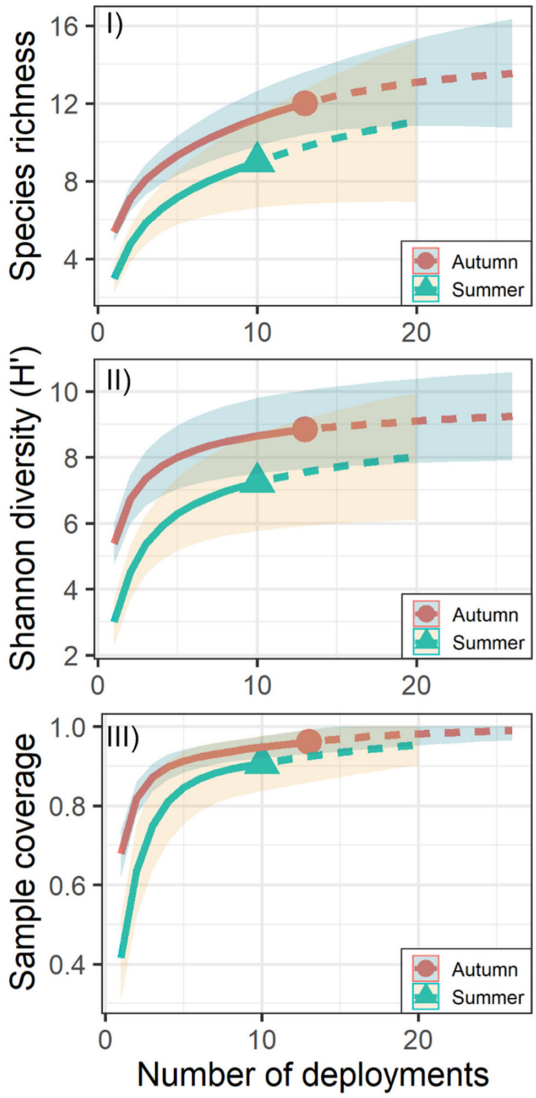

FIGURE 2 Sample size-based interpolation (solid lines) and extrapolation (dashed lines) curves for cumulative species richness (I), diversity (II) and sample coverage (III), with 95% confidence bands (shaded areas), for coastal rocky fish species in Baited Remote Underwater Video Stations (BRUVS) at Morro Gonzalo in the Los Ríos region of south-central Chile on 16–17 January (summer) and 29–30 April (autumn) 2022.

including the narrow-mouthed catshark (*Schroederichthys* sp.) and four commercially important species of rocky reef fish (*Graus nigra*, *Genypterus chilensis*, *Aplodactylus punctatus* and *Cheilodactylus variegatus*). Two of these taxa (*G. chilensis* and *A. punctatus*) were recorded only in the summer survey, and the other five (*S. viridis*, *G. nigra*, *H. cunninghami*, *Schroederichthys* sp. *and C. variegatus*) were recorded only in the autumn survey.

Average richness and diversity differed significantly between seasons, with higher values in autumn (Table S1, Figure 2). However, rarefaction curves showed that biodiversity metrics were similar between sampling seasons, considering the whole rocky reef fish assemblage of the study site, because 95% confidence intervals overlapped (Figure 2). Conversely, total abundance (MaxN) was qualitatively higher and TFA per sampling unit was qualitatively lower in autumn (i.e. not significantly different, Figure 4).

Species of the highest relative abundance, *P. jugularis*, *P. chilensis*, *S. oculatus*, *Patagonotothen* sp. and *Eptatretus* sp., were qualitatively higher in autumn surveys (Figure 5), but only *Eptatretus* sp. was significantly more abundant in autumn ($p < 0.05$, Table S2).

4 | DISCUSSION

We successfully used a single BRUVS camera to estimate relative abundance, species richness and diversity of a rocky reef fish assemblage in a coastal temperate site in central-southern Chile, as an example of an easy, fast and low-cost method of sampling natural environments to address the complexity of ecosystem services to humans, particularly in developing countries. Ecological factors, survey design effects or previous species-specific fishing impacts could have influenced our findings, although rare species (excluding *G. chilensis*) were on the southern limits of their distribution (Reyes & Hune, 2012; The Nature Conservancy, 2020). Further, seven species medium- to large-sized registered in this study represented a high proportion of nine species targeted by commercial fishing in the survey area (Reyes & Hune, 2012; The Nature Conservancy, 2020). Interviews with local fishers indicated that *Eleginops maclovinus* (Eleginopsidae) and *Sicyases sanguineus* (Gobiesocidae) were caught by commercial and recreational fisheries in the Los Ríos region (Canales et al., 2022), but were not recorded by BRUVS. The distribution of these two species in nearshore environments is strongly influenced by tide-related movements to feed in intertidal areas and aggregate near river mouths (Reyes & Hune, 2012), which make BRUVS (due the hard wave action, currents and turbidity inherent to these section of the coastal areas) or areas we sampled ineffective for these species. Although BRUVS are already established for sampling some fisheries management around the world (e.g. Thompson et al., 2022), a comparison of estimates from BRUVS with other independent sources of information are recommended to rule out the effect of other variables on estimates of relative abundance, richness and diversity.

Both of our sampling events were during the warm season of the year (summer and early autumn); thus, we did not detect expected a strong seasonal patterns for a temperate coastal area (i.e. Relative abundances, species richness and diversity variation at seasonal scale) (Irigoyen et al., 2013). Significantly higher abundance of *Eptatretus* sp. in autumn (recorded in ~90% of deployments) than summer (30% of deployments) was likely because the genus *Eptatretus* undergoes seasonal bathymetric migration (Li et al., 2022). Use of BRUVS also enabled observation of *G. chilensis* feeding during the day, although local fisher knowledge indicated that this species feeds nocturnally where commercial fishing takes place. Therefore, BRUVS can provide new information on nictemeral behaviour of rockfish species.

Chile has a relatively long history of fisheries monitoring and management, although fisheries for coastal rocky species in Chile

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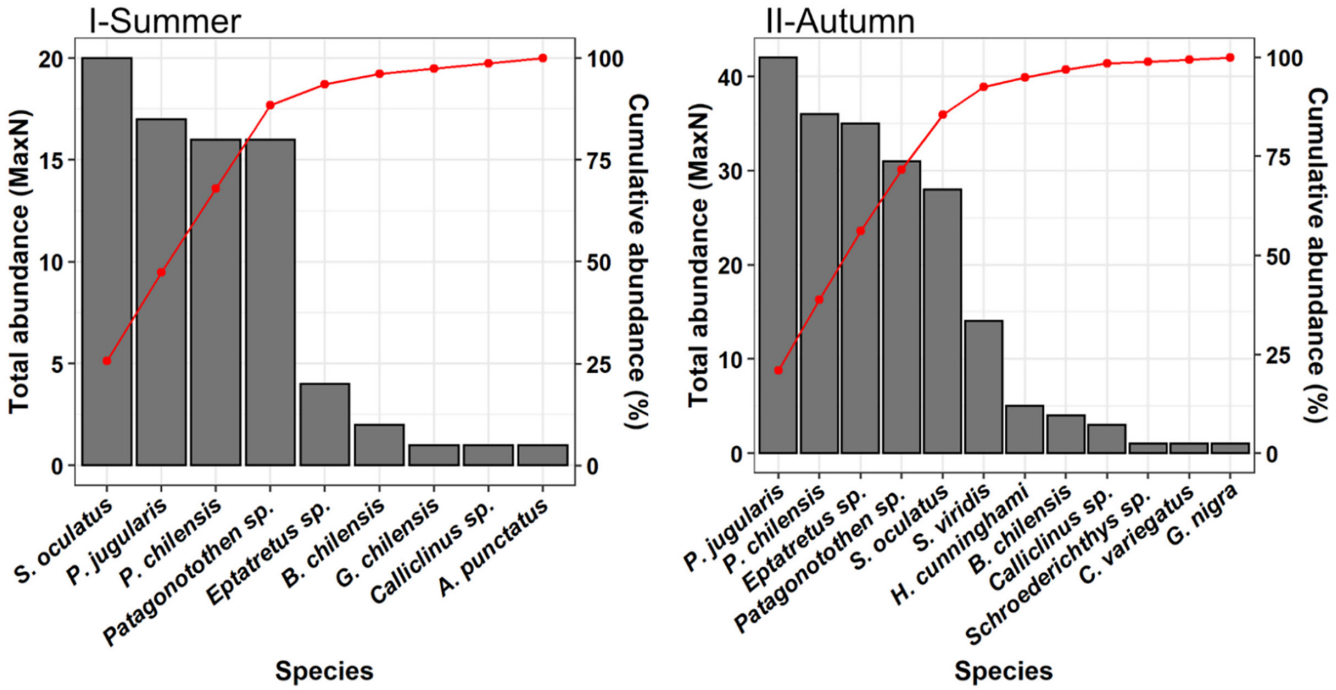

FIGURE 3 Total abundance (MaxN = bars) and cumulative abundance of coastal rocky fish species (lines) recorded in Baited Remote Underwater Video Stations (BRUVS) at Morro Gonzalo in the Los Ríos region of south-central Chile on 16–17 January (summer) and 29–30 April (autumn) 2022.

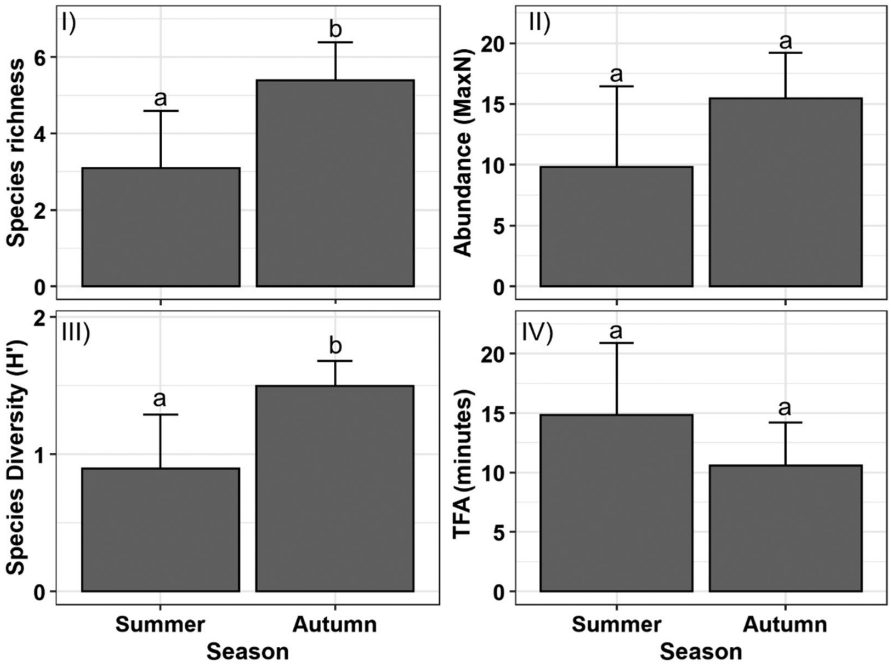

FIGURE 4 Mean species richness (I), abundance (II = MaxN), Shannon diversity (III) and time of first arrival (IV = TFA) per deployment (error bars = 1 SD) of coastal rocky fish species in Baited Remote Underwater Video Stations (BRUVS) at Morro Gonzalo in the Los Ríos region of south-central Chile on 16–17 January (summer) and 29–30 April (autumn) 2022. Lowercase letters identify significant differences between summer and autumn (*p*-values adjusted with the Bonferroni correction).

are labelled as "open access regime", for which registration of new fishers is allowed, fishing is not monitored, and stock status is not assessed. Management is limited to fishing gear type restrictions for most rockfish species by the Undersecretary of Fisheries and Aquaculture (SUBPESCA, decreed No. 3916/2005). However, some of the coastal rocky species targeted by commercial and recreational fisheries in Chile have started to show signs of depletion (Godoy et al., 2010). Signs of depletion should trigger a reclassification of these fisheries currently under a fully exploited regime, for which monitoring should be established and exploitation status should be evaluated using maximum sustainable yield (MSY) based reference points (Wiff et al., 2016). Although rocky fish species in Chile may not be classified as fully exploited for years or decades, some management actions are currently needed. Effort restriction or quota

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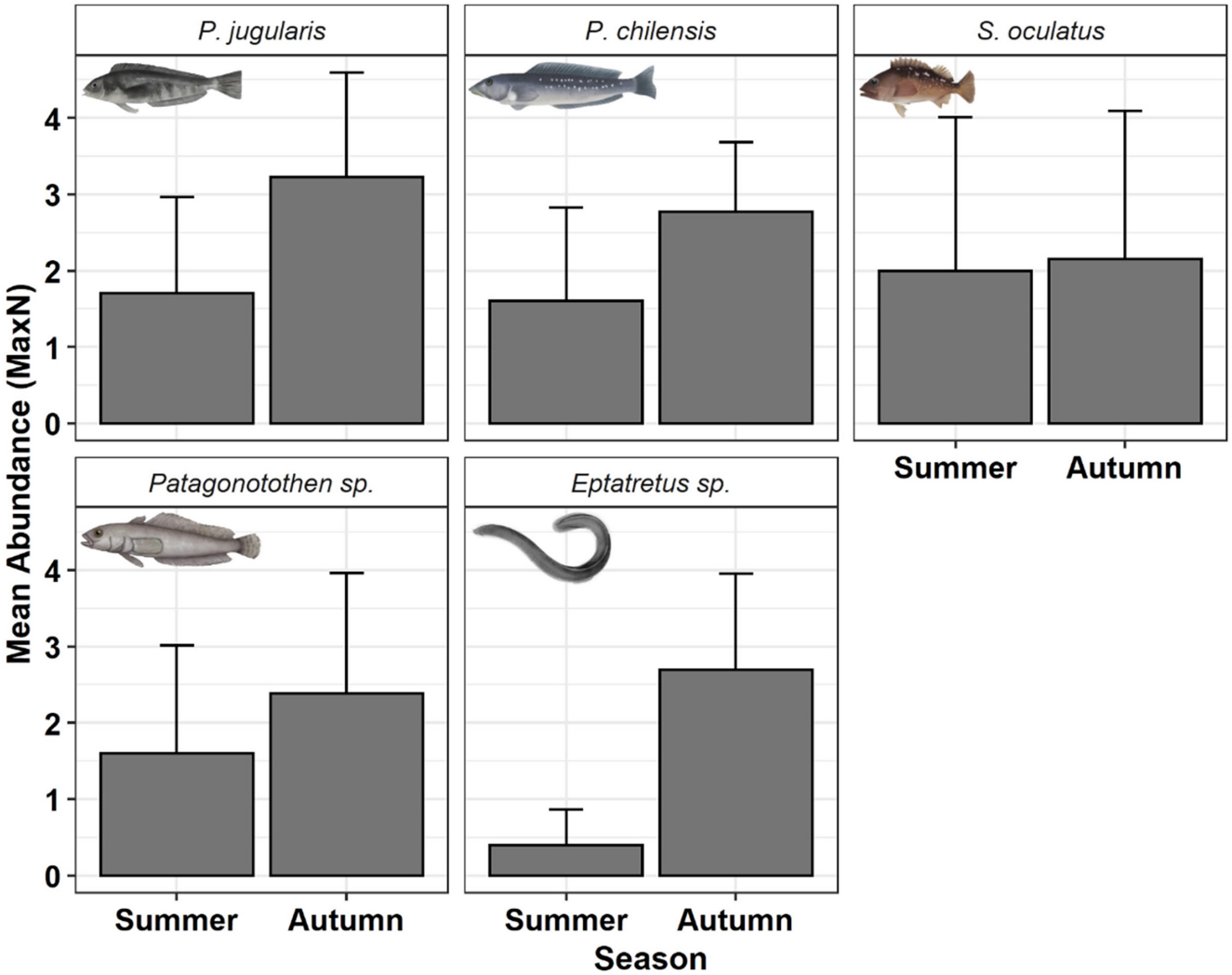

FIGURE 5 Mean abundance (MaxN) per deployment (±SD) of the five most conspicuous rocky fish species in Baited Remote Underwater Video Stations (BRUVS) at Morro Gonzalo in the Los Ríos region of south-central Chile on 16–17 January (summer) and 29–30 April (autumn) 2022. Fish images are from "PECES COSTEROS DE CHILE: GUÍA DE IDENTIFICACIÓN DE CAMPO – Nature Conservancy" and from www.proyectoarrecife.com.ar).

application is not in place for coastal rocky species because of a lack of basic data on catches and relative abundance.

By collecting data on size structure of fish species and using a BRUVS-derived abundance index, MSY can be easily estimated to comply with Chilean fishing law. When estimating an abundance index, detectability of species to BRUVS should be maintained constant within each fishing location, by considering bottom type, visibility and depth as key factors or influencing variables. Such factors may be fixed, weighted or used as covariates in the analysis of different data sources or sampling areas (e.g. Thompson et al., 2022). In this context, other BRUVS survey attributes, such as the time of year (e.g. warm or cold season) and type and size of bait, should be maintained as consistently as possible among surveys. In addition, other sources of bias related to life history of fish species must be considered in case of further development and application of BRUVs. For example, relative abundance is expected to be underestimated for nocturnal species (e.g. *G. chilensis*), or herbivorous species not attracted by fish bait. In our study, *Aplodactylus punctatus* and *Scartichthys viridisis* were the only herbivorous species described in the region (Reyes & Hune, 2012), thus, their abundances were most likely underestimated. In addition to relative abundance trends of commercial species, medium- to long-term monitoring of coastal areas by BRUVS may help to detect changes in fish assemblages associated with environmental change. Furthermore, video imagery provides a permanent record that can be re-analysed for purposes other than those originally planned.

We used single-camera BRUVS, the width and size of which allowed deployment by hand from small boats, even in swell and windy conditions. In more favourable weather conditions, sampling can take just a single working day, which makes BRUVS a cost-efficient sampling tool. However, single-camera BRUVS (in comparison to stereo-BRUVS) do not enable collection of detailed information on size structure, although other low-cost techniques to estimate fish length may be available (Richardson et al., 2015). In particular,

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BRUVS equipped with SJCAM action cameras were assembled at a cost of less than US$300, an order of magnitude lower than standard stereo-BRUVS equipment used in developed countries. Furthermore, manual deployment of BRUVS from small boats was a much lower daily sampling cost than using heavy equipment from large boats or research vessels. The simple, inexpensive and light single-camera BRUVS we used may be a key factor in maintaining long-term monitoring programmes along remote coastal areas of Chile.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

Information on ethical treatment of humans and animals is not applicable as only video records on fishes were used for data gathering. The manuscript has not been submitted for publication or published in another journal.

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